

B.TECH 1ST YEAR · 1ST SEMESTER · ENGINEERING MATHEMATICS

Vector Integration

Unit V — Complete Formula Sheet

Last-Day Revision · Exam-Ready · All Formulas + Theorems + Flashcards



LINE INTEGRAL

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$



SURFACE INTEGRAL

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$$



VOLUME INTEGRAL

$$\iiint_V \phi \, dV$$



GREEN'S THEOREM

Line → **Area**



STOKES' THEOREM

Line ↔ **Surface**



DIVERGENCE THM.

Volume ↔ **Surface**

✓ No long derivations | ✓ Only key formulas & results | ✓ Flashcards included | ✓ Exam tips & quick revision sheet



⚙️ Course Objectives

- 01 Understand line, surface, and volume integrals and their physical meanings.
- 02 Apply circulation and work done concepts using line integrals.
- 03 Evaluate flux across a surface using surface integrals.
- 04 Use Green's theorem to convert line integrals to area integrals.
- 05 Apply Stokes' theorem relating line and surface integrals.
- 06 Use the Divergence (Gauss) theorem relating volume and surface integrals.

📖 Basic Definitions

📌 Vector & Scalar Fields

Vector Field $\mathbf{F}(x, y, z)$

A function that assigns a vector to every point in space.

$$\mathbf{F} = P \hat{i} + Q \hat{j} + R \hat{k}$$

Scalar Field $\phi(x, y, z)$

A function that assigns a scalar value to every point in space.

📌 Differential Elements

$$d\mathbf{r} = dx \hat{i} + dy \hat{j} + dz \hat{k} \quad (\text{line element})$$

$$d\mathbf{S} = \hat{n} \, dS \quad (\text{vector surface element})$$

$$dV = dx \, dy \, dz \quad (\text{volume element})$$

📌 Short Definitions

Line Integral

Integral of a field along a curve C in space.

Circulation

Line integral of a vector field around a **closed** loop — measures "rotation."

Work Done

Work = $\int_C \mathbf{F} \cdot d\mathbf{r}$ — total work done by force $\mathbf{F}$ along path C .

Flux

Rate of flow of a vector field **through** a surface — measures how much field passes through S .

Unit Normal $\hat{n}$

A unit vector perpendicular to the surface at each point.

📌 Types of Integrals — At a Glance

Type	Symbol	Formula	Dimension	Physical Meaning
Line Integral	$\int_C$	$\int_C \mathbf{F} \cdot d\mathbf{r}$	1D (curve)	Work done / circulation
Surface Integral	$\iint_S$	$\iint_S \mathbf{F} \cdot \hat{n} \, dS$	2D (surface)	Flux through surface
Volume Integral	$\iiint_V$	$\iiint_V \phi \, dV$	3D (volume)	Total quantity in volume

Line Integral — Circulation & Work Done

Definition

Integral of a vector field $\mathbf{F}$ along a curve C from point A to B .

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (P \, dx + Q \, dy + R \, dz)$$

Work Done

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}$$

Total work done by force $\mathbf{F}$ moving along path C .

Circulation

$$\Gamma = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

Line integral along a **closed path** — measures rotation of the field.

Parametric Form

If $C: \mathbf{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$, $t \in [a, b]$:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$$

Key Notes

- ✓ Line integral is generally **path-dependent**.
- ✓ If $\mathbf{F}$ is conservative: **path-independent** and $\oint \mathbf{F} \cdot d\mathbf{r} = 0$.
- ✓ Closed path uses symbol $\oint$.
- ✓ For $\mathbf{F} = \nabla\phi$: $\int_C \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A)$.
- ✓ $\mathbf{F}$ is conservative $\Leftrightarrow \nabla \times \mathbf{F} = \mathbf{0}$.

Surface Integral — Flux

Definition & Formula

Integral of vector field $\mathbf{F}$ over surface S — measures total flux.

$$\iint_S \mathbf{F} \cdot \hat{n} \, dS = \iint_S \mathbf{F} \cdot d\mathbf{S}$$

$$\text{Flux} = \iint_S \mathbf{F} \cdot \hat{n} \, dS$$

Unit Normal Vector $\hat{n}$

$$\hat{n} = \nabla\phi / |\nabla\phi|$$

For $z = f(x, y)$: $\hat{n} = (-f_x \hat{i} - f_y \hat{j} + \hat{k}) / \sqrt{f_x^2 + f_y^2 + 1}$

Evaluation Formula

If surface $S: z = f(x, y)$, projected onto R in xy -plane:

$$\iint_S \mathbf{F} \cdot \hat{n} \, dS = \iint_R \mathbf{F} \cdot (-f_x \hat{i} - f_y \hat{j} + \hat{k}) \, dA$$

Types of Surfaces

Type	Property
Open Surface	Has a boundary curve (e.g., hemisphere)
Closed Surface	No boundary — encloses a volume (e.g., sphere)
Orientable	Has consistent normal direction throughout

Volume Integral

Definition & Formula

Integral of scalar field ϕ over a 3D region V .

$$\iiint_V \phi(x, y, z) \, dV$$

$$= \iiint \phi \, dx \, dy \, dz$$

Physical Meaning: Total mass, charge, or any scalar quantity contained in volume V (when ϕ is density).

Common Forms

$$\iiint_V \nabla \cdot \mathbf{F} \, dV = \iint_S \mathbf{F} \cdot \hat{n} \, dS \quad (\text{Divergence Thm.})$$

$$\iiint_V 1 \, dV = \text{Volume of } V$$

Key Notes

- ✓ Limits depend on geometry of region V .
- ✓ Use cylindrical / spherical coordinates for symmetric regions.
- ✓ Cylindrical: $dV = r \, dr \, d\theta \, dz$
- ✓ Spherical: $dV = r^2 \sin \phi \, dr \, d\theta \, d\phi$

⚡ Green's Theorem (Plane) — Without Proof

● Statement of Green's Theorem

If $M(x,y)$ and $N(x,y)$ have continuous partial derivatives in a region R bounded by a simple closed curve C (traversed counterclockwise), then:

$$\oint_C (M \, dx + N \, dy) = \iint_R (\partial N / \partial x - \partial M / \partial y) \, dA$$

★ Conditions

- ✓ M and N must have continuous first partial derivatives in R .
- ✓ C is a simple, closed, positively oriented (counterclockwise) curve.
- ✓ R is the region enclosed by C .
- ✓ C must not cross itself.

★ Region & Boundary Relation

∂R = C (Boundary)

C is the boundary of region R . Positive orientation = counterclockwise traversal so R is always on the LEFT.

★ Applications

1. Area Calculation

$$A = \frac{1}{2} \oint_C (x \, dy - y \, dx)$$

$$A = \oint_C x \, dy = -\oint_C y \, dx$$

2. Line → Double Integral Conversion

Converts a difficult closed-curve line integral to an easier double integral over the enclosed region R .

★ Vector Form of Green's Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{k}} \, dA$$

Green's theorem is a 2D special case of Stokes' theorem (in xy -plane).

Symbol	Meaning
$\oint_C$	Closed line integral (counterclockwise)
$\iint_R$	Double integral over region R
$\partial N / \partial x - \partial M / \partial y$	Curl in z -direction
$dA = dx \, dy$	Area element of R

★ Quick Formula Summary

Purpose	Formula
Standard form	$\oint_C (M \, dx + N \, dy) = \iint_R (\partial N / \partial x - \partial M / \partial y) \, dA$
Area of region R	$A = \frac{1}{2} \oint_C (x \, dy - y \, dx)$
Conservative field check	$\partial N / \partial x = \partial M / \partial y \rightarrow \text{integral} = 0$

⚠ **Remember:** Green's theorem works in the **plane** (2D) only. Curve C must be traversed **counterclockwise** (positive orientation). If clockwise, add a minus sign.

Stokes' Theorem — Without Proof

Statement of Stokes' Theorem

If $\mathbf{F}$ is a vector field with continuous partial derivatives on an open, oriented surface S with boundary curve C (oriented consistently with S), then:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} \, dS$$

Key Relation

Left Side	Right Side
$\oint_C \mathbf{F} \cdot d\mathbf{r}$	$\iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} \, dS$
Circulation around boundary curve C	Flux of curl of $\mathbf{F}$ through surface S

Curl Formula

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix}$$

$$= (R_y - Q_z)\hat{\mathbf{i}} - (R_x - P_z)\hat{\mathbf{j}} + (Q_x - P_y)\hat{\mathbf{k}}$$

Conditions

- ✓ $\mathbf{F}$ has continuous partial derivatives on S .
- ✓ S is an open, smooth, orientable surface.
- ✓ $C = \partial S$ is the positively oriented boundary of S .
- ✓ Right-hand rule: curl fingers along C , thumb points in direction of $\hat{\mathbf{n}}$.

Applications

- ✓ Evaluate difficult line integrals using surface integrals.
- ✓ Evaluate surface integrals by computing a line integral instead.
- ✓ Prove that $\oint \mathbf{F} \cdot d\mathbf{r} = 0$ when $\nabla \times \mathbf{F} = 0$ (irrotational field).
- ✓ Relates microscopic rotation (curl) to macroscopic circulation.

Symbol Reference

Symbol	Name	Meaning
$\oint_C \mathbf{F} \cdot d\mathbf{r}$	Circulation	Line integral around closed boundary C
$\nabla \times \mathbf{F}$	Curl of $\mathbf{F}$	Rotation/vorticity of the vector field
$\hat{\mathbf{n}}$	Unit normal	Outward/upward normal to surface S
$\iint_S$	Surface integral	Integral over open surface S
$\partial S = C$	Boundary	C is the boundary curve of S

Special Cases

Green's Theorem as Special Case

When S lies in the xy -plane and $\hat{\mathbf{n}} = \hat{\mathbf{k}}$:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{k}} \, dA$$

This is exactly Green's theorem!

Irrotational Field

If $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere in S , then:

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$$

Field is conservative (irrotational).

Remember: Stokes' theorem works in **3D** with an **open surface**. Always verify the orientation using the **right-hand rule**. Green's theorem is the 2D version of Stokes' theorem.

■ Divergence Theorem (Gauss's Theorem) — Without Proof

● Statement of Divergence Theorem

If $\mathbf{F}$ is a vector field with continuous partial derivatives in a volume V enclosed by a closed surface S , then:

$$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$$

★ Divergence Formula

$$\nabla \cdot \mathbf{F} = \partial P / \partial x + \partial Q / \partial y + \partial R / \partial z$$

where $\mathbf{F} = P\hat{i} + Q\hat{j} + R\hat{k}$

★ Physical Meaning

Left Side: $\iiint_V (\nabla \cdot \mathbf{F}) \, dV$

Total "generation" (sources – sinks) of the field inside volume V .

Right Side: $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$

Total outward flux through the closed surface S enclosing V .

Net outward flux = Total divergence inside

★ Conditions

- ✓ $\mathbf{F}$ has continuous first partial derivatives in V .
- ✓ S is a closed, piecewise-smooth surface.
- ✓ S completely encloses volume V .
- ✓ $\hat{\mathbf{n}}$ is the **outward** unit normal on S .

★ Applications

- ✓ Convert difficult surface integrals into volume integrals.
- ✓ Compute flux through complex closed surfaces.
- ✓ Prove that flux = 0 for divergence-free (solenoidal) fields.
- ✓ Fluid mechanics: net flow out = rate of fluid created inside.
- ✓ Electrostatics: Gauss's law derivation.

★ Key Symbol Reference

Symbol	Name	Meaning
$\iiint_V (\nabla \cdot \mathbf{F}) \, dV$	Volume Integral of Divergence	Total source/sink strength in V
$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$	Flux Integral	Total outward flux through closed surface S
$\nabla \cdot \mathbf{F}$	Divergence	Outward tendency of the field at a point
$\hat{\mathbf{n}}$	Outward Normal	Must point OUTWARD for the theorem to hold
$\partial S = S$	Boundary	S is the closed boundary of V

★ Special Case: Solenoidal Field

If $\nabla \cdot \mathbf{F} = 0$ (Solenoidal)

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS = 0$$

Total flux through any closed surface = 0.

Volume of region V

$$V = \frac{1}{3} \iint_S (x n_x + y n_y + z n_z) \, dS$$

Using Divergence theorem with $\mathbf{F} = \mathbf{r}$.

⚠ Remember: Divergence theorem requires a **CLOSED surface** with $\hat{\mathbf{n}}$ pointing **outward**. Do NOT confuse with Stokes' theorem (open surface). If $\nabla \cdot \mathbf{F} = 0$, the flux is automatically zero.

📄 Relation Between Theorems — Master Comparison

Feature	Green's Theorem	Stokes' Theorem	Divergence Theorem
Also Known As	Green's theorem in the plane	Curl theorem	Gauss's theorem
Formula	$\oint_C (M \, dx + N \, dy) = \iint_R (\partial N / \partial x - \partial M / \partial y) \, dA$	$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} \, dS$	$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$
Dimension	2D (plane)	3D (surface)	3D (volume)
LHS Type	Closed line integral $\oint$	Closed line integral $\oint$	Volume integral $\iiint$
RHS Type	Double integral $\iint$ over area R	Surface integral $\iint$ over surface S	Surface integral $\iint$ over closed surface S
Operator Used	$\partial N / \partial x - \partial M / \partial y$ (2D curl)	$\nabla \times \mathbf{F}$ (3D curl)	$\nabla \cdot \mathbf{F}$ (divergence)
Surface Type	Flat region R in xy-plane	Open surface S in 3D	Closed surface S in 3D
Boundary	Closed curve $C = \partial R$	Closed curve $C = \partial S$	Closed surface $S = \partial V$
Normal $\hat{\mathbf{n}}$ direction	$\hat{\mathbf{k}}$ (upward, out of plane)	Consistent with right-hand rule	Outward from volume V
Physical Meaning	Circulation = area integral of curl	Circulation = flux of curl	Outward flux = total divergence inside
Special Case	Green's = 2D Stokes'	Reduces to Green's in xy-plane	Flux = 0 if $\nabla \cdot \mathbf{F} = 0$
Zero Condition	$\partial N / \partial x = \partial M / \partial y \rightarrow \oint = 0$	$\nabla \times \mathbf{F} = 0 \rightarrow \oint = 0$	$\nabla \cdot \mathbf{F} = 0 \rightarrow \iint = 0$

📌 Circulation vs Flux

Property	Circulation	Flux
Formula	$\oint_C \mathbf{F} \cdot d\mathbf{r}$	$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$
Integration over	Closed curve C	Surface S
Measures	Rotation / vorticity	Flow through surface
Related operator	Curl: $\nabla \times \mathbf{F}$	Divergence: $\nabla \cdot \mathbf{F}$
Theorem used	Stokes / Green	Divergence (Gauss)
Zero when	Field is irrotational	Field is solenoidal

📌 Open vs Closed Surfaces

	Open Surface	Closed Surface
Definition	Has a boundary curve	Encloses a volume, no boundary
Example	Hemisphere, paraboloid cap, disk	Sphere, cube, cylinder (closed ends)
Normal direction	Consistent with right-hand rule	Outward (away from enclosed volume)
Theorem applied	Stokes' theorem	Divergence theorem
Boundary	$\partial S =$ closed curve C	$\partial V = S$ (no curve boundary)

📌 Theorem Applications Summary

Theorem	Use When	Benefit
Green's	Have closed curve in xy-plane	Convert line integral $\rightarrow$ area integral
Stokes'	Have closed curve in 3D space	Convert line integral $\leftrightarrow$ surface integral
Divergence	Have closed surface in 3D	Convert surface integral $\leftrightarrow$ volume integral

Flashcards — Section A: Definitions & Basic Formulas (1–15)

Q1: What is a line integral?

Integral of a field along a curve C in space.

Q3: Work done formula?

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}$$

Q5: Surface integral formula?

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$$

Q7: Volume integral formula?

$$\iiint_V \varphi(x, y, z) \, dV$$

Q9: What is unit normal $\hat{\mathbf{n}}$?

A unit vector perpendicular to surface S at each point.

Q11: When is field conservative?

When $\nabla \times \mathbf{F} = \mathbf{0}$ (irrotational field).

Q13: Volume element in cylindrical coords?

$$dV = r \, dr \, d\theta \, dz$$

Q15: What does $\oint$ symbol mean?

Closed line/surface integral (along closed path/surface).

Q2: Line integral formula?

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (P \, dx + Q \, dy + R \, dz)$$

Q4: Circulation formula?

$$\Gamma = \oint_C \mathbf{F} \cdot d\mathbf{r} \text{ (closed path)}$$

Q6: Flux represents?

Total flow of field $\mathbf{F}$ through surface S .

Q8: Physical meaning of volume integral?

Total mass/charge/quantity inside volume V .

Q10: Parametric form of line integral?

$$\int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$$

Q12: For conservative field $\mathbf{F} = \nabla\varphi$:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \varphi(B) - \varphi(A)$$

Q14: Volume element in spherical coords?

$$dV = r^2 \sin \varphi \, dr \, d\theta \, d\varphi$$

Flashcards — Section B: Theorems (16–30)

Q16: Green's theorem converts?

Line integral $\rightarrow$ Area (double) integral over R .

Q18: Green's theorem: area formula?

$$A = \frac{1}{2} \oint_C (x \, dy - y \, dx)$$

Q20: Stokes' theorem relates?

Line integral (circulation) $\leftrightarrow$ Surface integral (flux of curl).

Q22: Stokes' theorem: surface type?

Open surface S with boundary curve C .

Q24: Divergence theorem relates?

Volume integral $\leftrightarrow$ Surface integral (flux).

Q26: Divergence theorem: surface type?

Closed surface S enclosing volume V .

Q28: If $\nabla \cdot \mathbf{F} = 0$, field is?

Solenoidal $\rightarrow$ total flux = 0.

Q30: Normal direction in Divergence theorem?

Outward normal (pointing away from enclosed volume V).

Q17: Green's theorem formula?

$$\oint_C (M \, dx + N \, dy) = \iint_R (\partial N / \partial x - \partial M / \partial y) \, dA$$

Q19: Orientation in Green's theorem?

Counterclockwise (positive orientation).

Q21: Stokes' theorem formula?

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} \, dS$$

Q23: Curl formula components?

$$\nabla \times \mathbf{F} = (R_y - Q_z)\hat{\mathbf{i}} - (R_x - P_z)\hat{\mathbf{j}} + (Q_x - P_y)\hat{\mathbf{k}}$$

Q25: Divergence theorem formula?

$$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$$

Q27: Divergence formula?

$$\nabla \cdot \mathbf{F} = \partial P / \partial x + \partial Q / \partial y + \partial R / \partial z$$

Q29: Green's theorem is which special case?

2D special case of Stokes' theorem ($\hat{\mathbf{n}} = \hat{\mathbf{k}}$).

⚡ Exam Tips — Common Mistakes to Avoid

✗ Mistake 1: Confusing Line and Surface Integrals

Line integral → along a **curve**: $\int_C \mathbf{F} \cdot d\mathbf{r}$

Surface integral → over a **surface**: $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$

Check dimension: 1D curve vs 2D surface.

✗ Mistake 2: Missing Unit Normal Vector $\hat{\mathbf{n}}$ in Flux

Flux = $\iint \mathbf{F} \cdot \hat{\mathbf{n}} \, dS$ — $\hat{\mathbf{n}}$ is mandatory!

$\hat{\mathbf{n}} = \nabla\phi / |\nabla\phi|$ for implicit surface $\phi = c$

Never write $\iint \mathbf{F} \, dS$ — always include $\hat{\mathbf{n}}$.

✗ Mistake 3: Sign Errors in Green's Theorem

Correct: $\partial N / \partial x - \partial M / \partial y$ (N first, M second!)

Wrong order: $\partial M / \partial x - \partial N / \partial y \leftarrow$ WRONG

M is coefficient of dx, N is coefficient of dy.

✗ Mistake 4: Orientation Errors

Green's theorem: C must be **counterclockwise**

Stokes' theorem: use **right-hand rule** for $\hat{\mathbf{n}}$

Divergence theorem: $\hat{\mathbf{n}}$ must point **outward**

Clockwise orientation → add minus sign!

✗ Mistake 5: Using Closed vs Open Surface

Stokes' theorem → **OPEN** surface (has boundary curve)

Divergence theorem → **CLOSED** surface (encloses volume)

Check if surface has a boundary or not!

✗ Mistake 6: Mixing Stokes and Divergence Theorems

Stokes: $\oint_C \mathbf{F} \cdot d\mathbf{r} \leftrightarrow \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} \, dS \rightarrow$ uses **curl**

Divergence: $\iiint_V (\nabla \cdot \mathbf{F}) \, dV \leftrightarrow \iint_S \mathbf{F} \cdot \hat{\mathbf{n}} \, dS \rightarrow$ uses **div**

Stokes = curl; Divergence = div. Never swap!

✗ Mistake 7: Errors in Limits of Integration

- Identify the region of integration carefully.

- Sketch the region before writing limits.

- For volume integrals, get z-limits from surface equations.

Wrong limits = wrong answer even with correct formula!

✗ Mistake 8: Forgetting to Check Conditions

- Green's: partial derivatives must be continuous in R

- Stokes': surface must be smooth and orientable

- Divergence: $\mathbf{F}$ must have continuous partial derivatives in V

Theorems fail if conditions are not satisfied!

💡 Golden Rules for Exams

- Always write the theorem statement before applying it.
- Verify conditions before using a theorem.
- State the orientation (CCW / outward normal).
- Sketch the curve/surface/region for clarity.
- For Green's: always check $\partial N / \partial x - \partial M / \partial y$ first.
- Closed curve $\oint \rightarrow$ think Green's or Stokes.
- Closed surface $\iint \rightarrow$ think Divergence theorem.
- If $\nabla \times \mathbf{F} = 0$: conservative $\rightarrow$ use potential function.
- Use symmetry to simplify calculations where possible.

🚀 Quick Revision Sheet — Formulas Only

∫ LINE INTEGRAL

BASIC FORM

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C (P dx + Q dy + R dz)$$

PARAMETRIC FORM

$$= \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

WORK DONE

$$W = \int_C \mathbf{F} \cdot d\mathbf{r}$$

CIRCULATION (CLOSED)

$$\Gamma = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

CONSERVATIVE FIELD

$$\mathbf{F} = \nabla\phi \rightarrow \int \mathbf{F} \cdot d\mathbf{r} = \phi(B) - \phi(A)$$

∬ SURFACE INTEGRAL (FLUX)

FLUX FORMULA

$$\Phi = \iint_S \mathbf{F} \cdot \hat{n} dS$$

UNIT NORMAL

$$\hat{n} = \nabla\phi / |\nabla\phi|$$

FOR Z = F(X,Y)

$$\iint_S \mathbf{F} \cdot \hat{n} dS = \iint_R \mathbf{F} \cdot (-f_x \hat{i} - f_y \hat{j} + \hat{k}) dA$$

∭ VOLUME INTEGRAL

BASIC FORM

$$\iiint_V \phi(x, y, z) dV = \iiint \phi dx dy dz$$

CYLINDRICAL

$$dV = r dr d\theta dz$$

SPHERICAL

$$dV = r^2 \sin\phi dr d\theta d\phi$$

● GREEN'S THEOREM

STATEMENT

$$\oint_C (M dx + N dy) = \iint_R (\partial N / \partial x - \partial M / \partial y) dA$$

AREA FORMULA

$$A = \frac{1}{2} \oint_C (x dy - y dx)$$

KEY: 2D | CURVE → AREA | CCW ORIENTATION

Converts: $\oint$ line → $\iint$ double integral

● STOKES' THEOREM

STATEMENT

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n} dS$$

CURL OPERATOR

$$\nabla \times \mathbf{F} = (R_y - Q_z)\hat{i} - (R_x - P_z)\hat{j} + (Q_x - P_y)\hat{k}$$

KEY: 3D | OPEN SURFACE | RIGHT-HAND RULE

Converts: $\oint$ line ↔ $\iint$ surface (curl)

● DIVERGENCE THEOREM

STATEMENT

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \iint_S \mathbf{F} \cdot \hat{n} dS$$

DIVERGENCE

$$\nabla \cdot \mathbf{F} = \partial P / \partial x + \partial Q / \partial y + \partial R / \partial z$$

KEY: 3D | CLOSED SURFACE | OUTWARD NORMAL

Converts: $\iiint$ volume ↔ $\iint$ surface (flux)

🔑 KEY IDENTITIES

IRROTATIONAL FIELD

$$\nabla \times \mathbf{F} = \mathbf{0} \Leftrightarrow \mathbf{F} \text{ conservative}$$

SOLENOIDAL FIELD

$$\nabla \cdot \mathbf{F} = \mathbf{0} \Leftrightarrow \text{flux} = 0$$

IDENTITY

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0 \text{ always}$$

IDENTITY

$$\nabla \times (\nabla \phi) = \mathbf{0} \text{ always}$$

🔑 Key Vector Identities & Operator Summary

Del Operator ∇

$$\nabla = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

Gradient, Divergence, Curl

Operation	Formula	Result
Gradient	$\nabla\varphi$	Vector field
Divergence	$\nabla \cdot \mathbf{F}$	Scalar field
Curl	$\nabla \times \mathbf{F}$	Vector field
Laplacian	$\nabla^2\varphi = \nabla \cdot (\nabla\varphi)$	Scalar field

Key Identities

$$\nabla \cdot (\nabla \times \mathbf{F}) = 0 \quad (\text{always})$$

$$\nabla \times (\nabla\varphi) = 0 \quad (\text{always})$$

$$\nabla \cdot (\varphi \mathbf{F}) = \varphi(\nabla \cdot \mathbf{F}) + \mathbf{F} \cdot (\nabla\varphi)$$

$$\nabla \times (\varphi \mathbf{F}) = \varphi(\nabla \times \mathbf{F}) + (\nabla\varphi) \times \mathbf{F}$$

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G})$$

Curl Expanded Form

$$\begin{aligned} \nabla \times \mathbf{F} = & (\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z})\hat{i} \\ & - (\frac{\partial R}{\partial x} - \frac{\partial P}{\partial z})\hat{j} \\ & + (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y})\hat{k} \end{aligned}$$

Divergence Expanded

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

Important Results

Condition	Name	Result
$\nabla \times \mathbf{F} = 0$	Irrotational	$\mathbf{F} = \nabla\varphi$; $\oint \mathbf{F} \cdot d\mathbf{r} = 0$
$\nabla \cdot \mathbf{F} = 0$	Solenoidal	$\iint \mathbf{F} \cdot \hat{n} \, dS = 0$
$\nabla^2\varphi = 0$	Harmonic	φ satisfies Laplace eq.

Physical Interpretations

Divergence $\nabla \cdot \mathbf{F}$ at a point

Rate of outward flow — source (+) or sink (–) strength.

Curl $\nabla \times \mathbf{F}$ at a point

Rotation/vorticity of field — angular velocity of fluid.

★ Final Summary — The Big Picture

Concept	Key Formula	Dimension	Physical Meaning	Theorem
Work Done	$\int_C \mathbf{F} \cdot d\mathbf{r}$	1D	Energy transferred by force	—
Circulation	$\oint_C \mathbf{F} \cdot d\mathbf{r}$	1D closed	Rotation of field	Green's / Stokes'
Flux	$\iint_S \mathbf{F} \cdot \hat{n} \, dS$	2D	Flow through surface	Divergence
Volume quantity	$\iiint_V \varphi \, dV$	3D	Total mass/charge in V	Divergence
Green's Thm.	$\oint (M \, dx + N \, dy) = \iint (N_x - M_y) \, dA$	2D plane	Circulation = area integral of curl	Line $\leftrightarrow$ Area
Stokes' Thm.	$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{n} \, dS$	3D open surface	Circulation = flux of curl	Line $\leftrightarrow$ Surface
Divergence Thm.	$\iiint_V (\nabla \cdot \mathbf{F}) \, dV = \iint_S \mathbf{F} \cdot \hat{n} \, dS$	3D closed surface	Outward flux = total divergence	Volume $\leftrightarrow$ Surface

🌀 **Remember:** Green's $\rightarrow$ 2D Plane | Stokes' $\rightarrow$ 3D Open Surface | Divergence $\rightarrow$ 3D Closed Surface
 Curl ($\nabla \times$) for circulation theorems · Divergence ($\nabla \cdot$) for flux theorem · CCW / Right-hand rule / Outward normal

